

Evaluation Criteria — L1-inception-epics-generator-agent

Quality Gates (must pass)

Criterion	Threshold	Method
All FRs covered	100% of input FRs in at least one feature's implements[]	Automated: set comparison
No orphan requirements	Zero FRs unassigned	Automated: count unassigned
Epic size 2-8 features	No violations	Automated: count per epic
IDs sequential	EPIC-01, EPIC-02... / FEAT-01-01, FEAT-01-02...	Automated: pattern check
No duplicate IDs	All IDs unique	Automated: uniqueness check
Priorities assigned	Every epic has a MoSCoW priority	Automated: field check
Features have implements[]	No empty implements	Automated: field check
Dependencies valid	All depends_on reference existing EPIC IDs	Automated: reference check

Evaluation Scores (LLM-as-Judge)

Evaluator	Threshold	Direction
Faithfulness	≥ 0.90	Every feature traces to FRs in input
Hallucination	≤ 0.10	No invented features beyond input requirements
Consistency	≥ 0.90	Epics don't overlap or contradict
Relevance	≥ 0.85	Groupings are logical and deliverable
Reasoning quality	≥ 0.80	Grouping and priority decisions explained
Citation completeness	≥ 0.95	Every feature cites FR IDs

Epic Quality Rubric

Dimension	Score 9-10	Score 7-8	Score 5-6	Score < 5
Grouping	Logical capability clusters, high cohesion	Mostly cohesive, one misplaced feature	Several features in wrong epic	Random grouping

Dimension	Score 9-10	Score 7-8	Score 5-6	Score < 5
Coverage	100% FR coverage, NFRs assigned	All FRs covered, NFRs partially	Some FRs missing	Major gaps
Sizing	All epics 3-8 features, right for 2-4 cycles	Most right-sized, one edge case	Several too large or too small	Epics undeliverable
Dependencies	Minimal, enable parallel delivery	Correct but serial where parallel possible	Unnecessary dependencies	Circular or missing
Traceability	Full FR → feature → epic chain	Mostly complete, minor gaps	Several broken links	No traceability

Reflection Checklist

The agent must self-verify before delivering:

- ☐ Every FR from input appears in at least one feature's implements[]
- ☐ NFRs are assigned as cross-cutting to relevant epics
- ☐ Constraints mapped to affected epics
- ☐ Assumptions mapped as risks
- ☐ No epic has only 1 feature (merge or decompose further)
- ☐ Dependencies are minimised — can any epics run in parallel?
- ☐ Epic priorities reflect the highest-priority FR they contain
- ☐ Execution summary includes coverage count and dependency chain

Reflection Process (mandatory)

The agent **MUST** perform reflection before delivering output:

1. **Generate** initial output following processing rules
2. **Log** [REFLECTING] Checking output against evaluation criteria
3. **Check** every item in the Reflection Checklist above
4. **Identify** gaps, errors, inconsistencies, or missed items
5. **Log** each finding: [REFLECTING] Found: <description>
6. **Fix** each issue — amend the output silently
7. **Log** each resolution: [REFLECTING] Resolved: <what was fixed>
8. **Deliver** only the final corrected output

The reflection findings and resolutions should appear in the execution_summary (what reflection

found and changed) but the interim output must never be shown.